Francis Qin

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EDUCATION

Northeastern University Boston, MA

Candidate for B.S in Electrical and Computer Engineering *Expected Apr 2028*

- *GPA*: 3.6/4.0
- *Relevant Coursework*: Circuits and Signals, Fundamentals of Engineering Algorithms, Fundamentals of Networks, Embedded Design/Enabling Robotics, Differential Equations & Linear Algebra, Physics II, Computing Fundamentals for Engineers, Databases, Discrete Structures, Fundamentals of Computer Science I & II
- *Activities*: ASME Student Member (contribute to hands-on design/build activities and technical workshops focused on mechanical design and prototyping), Men's Club Tennis Player (train in weekly practices and play in collegiate club tennis tournaments representing Northeastern University)
- *Dean's List*: Fall 2024, Spring 2025

SKILLS

Applications: SolidWorks, AutoCAD, Quartus Prime, LTspice, 3D Printing, Laser Cutting, MATLAB, Raspberry Pi Pico, DE1-SoC
Lab Equipment: Digital Multimeter, Oscilloscope, Function Generator
Operating Systems: Linux, Windows
Programming Languages: Python, Java, C++, Kotlin
Certifications: SolidWorks (CSWA)
Languages: Chinese (conversational)

WORK EXPERIENCES

Piotech Inc. | SolidWorks, PECVD/ALD Systems Shanghai, China

Electrical Engineering Intern *May 2026 - Aug 2026*

- Designed and iterated CAD models in SolidWorks, including an air compressor assembly, refining part geometry and assemblies across multiple design revisions to meet manufacturing tolerances
- Performed maintenance and troubleshooting on production machinery for PECVD and ALD thin film deposition systems used in semiconductor manufacturing, supporting equipment uptime on the production floor
- Collaborated with engineers and mentors to understand system-level operation of deposition tools, from process chambers to control subsystems, and applied that knowledge to daily troubleshooting
- Evaluated semiconductor equipment localization trends in China, including competitive positioning against NAURA, AMEC, and international suppliers, to contextualize the company's market strategy
- Documented equipment specifications, maintenance procedures, and testing results to support internal engineering records and knowledge transfer to future interns
- Shadowed cross-functional teams across manufacturing, quality control, and R&D to gain a broader understanding of semiconductor equipment production workflows

Chester Valley Country Club | Tennis Instruction, Client Coaching Chester County, PA

Tennis Coach *May 2025 - Aug 2025*

- Held private lessons and summer clinics for 6–13 year-old players, tailoring instruction by skill level and tracking progress

PROJECTS

Servo-Powered Robotic Arm | SolidWorks, Raspberry Pi Pico, 3D Printing May 2026 - Aug 2026

- Designed and iterated CAD models of a servo-powered robotic arm in SolidWorks, refining part geometry and assemblies across multiple design revisions
- 3D printed the full robotic arm assembly on a Bambu Lab printer, validating fit, range of motion, and structural integrity before final assembly
- Programmed a Raspberry Pi Pico to control servo motors driving the robotic arm, enabling coordinated joint movement and testing motion sequences
- Calibrated servo positioning and motion sequencing through iterative testing to improve precision and repeatability of arm movements
- Documented design iterations and testing outcomes to guide future improvements to the arm's mechanical and control systems

Game Design Project | SolidWorks, AutoCad, 3D Printing, Laser Cutting Oct 2025 – Dec 2025

Engineering Problem Solving and Computation

- Led end-to-end development of a fully functional tabletop game for two clients
- Produced complete CAD package (part models, assemblies, and manufacturing drawings) in SolidWorks/AutoCAD
- Ran iterative prototyping cycles using 3D printing and laser cutting, refining geometry and assembly based on playtesting feedback and failure points
- Drove design reviews within a 3-person team by documenting decisions, testing outcomes, and revisions to converge on a client-approved final prototype

Cryptocurrency Platform | Python, Streamlit, MySQL, Flask, REST API, Git, Docker Mar 2025 – Apr 2025

Introduction to Databases

- Built a full-stack cryptocurrency web app featuring user wallets, project views, and transaction workflows
- Implemented a Flask REST API with 50+ endpoints integrated with a MySQL database for dynamic queries and data operations
- Containerized services with Docker Compose for reproducible development and deployment alongside a Streamlit frontend